University
Mohammed VI
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College of Computing

Deliverable #: Lab-5/Normalization and SQL Implementation

Data Management Course

UM6P College of Computing

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1 -Introduction

This deliverable focuses on the normalization of the MNHS database schema and its implementation using SQL ,plus the implementation of some queries. The goal is to ensure data integrity, reduce redundancy, and optimize database performance.

2 -Requirements

In this deliverable, we are going to:

- Validate all the relations of the MNHS database schema against BCNF.
- Verify that all functional dependencies are preserved during normalization.
- Implement the normalized schema using SQL.
- Alter the table Patient and Staff to add new attributes (email and home adrress).
- Insert 5 rows of data into each table.
- Update the attributes phone number for the table patient and the attribute region for the attribute hospital.
- Delete a scheduled appointment from the table appointment.
- Write and execute SQL queries to retrieve specific information from the database.

3 -Methodology

- Query1: In order to have the second part of the full name (last name) we had to use a function called SUBSTRING_INDEX.
- Query2: We understand from The query "List distinct insurance types" we need to retrieve the different categories of insurance coverage stored in the TYPE attributes without showing any repeated values . This is achieved using the SQL keyword DISTINCT .
- Query3: To get the city and staff names we used a JOIN operation to combine data from the Staff,Hospital,Department and Workin tables based on the IDs.
- Query4: To find appointments scheduled for a specific interval, we used a WHERE clause to filter the results based on the appointment's Status and the function CURDATE() + INTERVAL 7 DAY.
- Query5: We used the function GROUP BY to group the results by department name and COUNT to count the number of appointments in each department.
- Query6: We used a JOIN operation to combine data from the Hospital and Stock tables. Then we used GROUP BY to calculate the average UnitPrice for each hospital.

- Query7: We used the COUNT function to calculate the total emergencies for every hospital.After that we used HAVING clause that filters the counted groups to show only those hospitals where the toatl count is greater than 20 .
- Query8: We first selected the medicaments that have the therapeutic class antibiotic ,then looked for those that cost less than 200 using the statment IN since the UnitPrice is in the Stock table.
- Query9: We used a normal WHERE clause to filter the UnitPrice for the medicaments that are the most expensive(normal comparison between the values within the table stock S and stock S2).
- Query10: This query was simillar to query 5 but instead of counting total appointments we calculated them based on their status (completed-scheduled-cancelled).And for that we used the CASE statement within the COUNT function.
- Query11: We used a JOIN operation to combine data from the Patient,Appointment and ClinicalActivity tables.Then we selected only the rows where the date of the next appointment is more than 30 days from the current date using the function CURDATE() + INTERVAL 30 DAY.
- Query12: We understand from the query 12 that we have two objectives:
  - First: Compute the total number of appointments .
  - Second: Compute the percentage share of appointments for the staff in his own hospital .

  To achieve this we used a subquery within the SELECT statements that calculate the total number of appointments for the entire hospital by linking it to the staff by the outer query using HID .Finally the percentage share is calculated using the formula Percentage_share =(staff_appointments_total / hospital_appointments_total)*100.
- Query13: For this query we tried to look for all the possible tuples that verify the conditions mentioned in the query .So we selected all the hospitals that have at least one drug that is out of stock.
- Query14: We first selected the IDs of that have all the antibiotics medicaments in stock.Then we used NOT EXISTS clause to find the hospitals that have all the antibiotics medicaments in stock.
- Query15: First we selected the average UnitPrice of the whole city hospitals using the function GROUP BY the TherapeuticClass .Then we joined this selection with the Medication and Stock tables to count the average by each hospital.Finally we used the clause CASE WHEN to flag the cases where the average UnitPrice in each hospital is higher or lower than the city average.
- Query16: We selected the minimum date of scheduled appointments of each patient using the function MIN along with GROUP BY PatientID.Then we joined this selection with the Patient,Appointment and ClinicalActivity tables to get their earliest appointment dates.

- Query17: We understand from the query 17 that we need to select patients based on two criteria applied to their emergency visits . First , we count the total number of emergency visits for each patient and use the having clause to filter only those whose counts are greater than 2 . After that we calculate the date of the latest visits and verify that this date falls within the last fourteen days . finally we group these two conditions using the logical operator AND.

- Query18: We first inner joined the Hospital,Department,ClinicalAcitivity and Appoinment tables to get all the necessary data .Then we grouped the results by hospital city and used COUNTto calculate how many completed appointmentseach hospital has .Finally we used RANK() window function to rank hospitals in each city based on the number of completed appointments in descending order in the last 90 days.

- Query19: For this query, we first used a JOIN operation to combine the Stock and Hospital tables in order to retrieve the medication ID (MID) along with the city it belongs to. Then, in the subquery, we grouped the Stock data by medication and city using GROUP BY, and we applied the functions MAX() and MIN() to calculate the price variation of each medication across hospitals. Finally, we filtered the results to include only those medications where the price variation exceeds 30 percent and used the IN statment to return only those medications from the main query whose price variability meets this condition.

- Query20: This query only checks the data quality by ensuring that the UnitPrice and Qty are both positive values.So we used a simple WHERE clause to filter any negative values within the Stock table.

4 -Implementation

4.1 -The refined MNHS Database Schema

```
    CREATE DATABASE MNHS;
USE MNHS;

CREATE TABLE Patient (
    IID INT PRIMARY KEY,
    CIN VARCHAR(10) UNIQUE NOT NULL,
    FullName VARCHAR(100) NOT NULL,
    Birth DATE,
    Sex ENUM('M', 'F') NOT NULL,
    BloodGroup ENUM('A+', 'A-', 'B+', 'B-', 'O+', 'O-',
       'AB+','AB-'),
    Phone VARCHAR(15)
);

CREATE TABLE Hospital (
    HID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    City VARCHAR(50) NOT NULL,
```

```
    Region VARCHAR(50)
);

CREATE TABLE Department (
    DEP_ID INT,
    HID INT,
    Name VARCHAR(100) NOT NULL,
    Specialty VARCHAR(100),
    PRIMARY KEY (DEP_ID),
    FOREIGN KEY (HID) REFERENCES Hospital(HID)
);

CREATE TABLE Staff (
    STAFF_ID INT PRIMARY KEY,
    FullName VARCHAR(100) NOT NULL,
    Status ENUM('Active', 'Retired') DEFAULT 'Active'
);

CREATE TABLE Workin (
    STAFF_ID INT,
    DEP_ID INT,
    primary key(STAFF_ID, DEP_ID),
    FOREIGN KEY (STAFF_ID) REFERENCES Staff(STAFF_ID),
    FOREIGN KEY (DEP_ID) REFERENCES Department(DEP_ID)
);

CREATE TABLE ClinicalActivity (
    CAID INT PRIMARY KEY,
    IID INT NOT NULL,
    STAFF_ID INT NOT NULL,
    DEP_ID INT NOT NULL,
    Date DATE NOT NULL,
    Time TIME,
    FOREIGN KEY (IID) REFERENCES Patient(IID),
    FOREIGN KEY (STAFF_ID) REFERENCES Staff(STAFF_ID),
    FOREIGN KEY (DEP_ID) REFERENCES Department(DEP_ID)
);

CREATE TABLE Appointment (
    CAID INT PRIMARY KEY,
    Reason VARCHAR(100),
    Status ENUM('Scheduled', 'Completed', 'Cancelled') DEFAULT '
       Scheduled',
    FOREIGN KEY (CAID) REFERENCES ClinicalActivity(CAID)
);

CREATE TABLE Emergency (
    CAID INT PRIMARY KEY,
    TriageLevel INT CHECK (TriageLevel BETWEEN 1 AND 5),
    Outcome ENUM('Discharged', 'Admitted', 'Transferred', '
       Deceased'),
```

```sql
    FOREIGN KEY (CAID) REFERENCES ClinicalActivity(CAID)
);

CREATE TABLE Insurance (
    INS_ID INT PRIMARY KEY,
    Type ENUM('CNOPS', 'CNSS', 'RAMED', 'Private', 'None') NOT
       NULL
);


CREATE TABLE Expense (
    EXP_ID INT PRIMARY KEY,
    INS_ID INT,
    CAID INT UNIQUE NOT NULL,
    Total DECIMAL(10, 2) NOT NULL CHECK (Total >= 0),
    FOREIGN KEY (INS_ID) REFERENCES Insurance(INS_ID),
    FOREIGN KEY (CAID) REFERENCES ClinicalActivity(CAID)
);

CREATE TABLE Medication (
    MID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Form VARCHAR(50),
    Strength VARCHAR(50),
    ActiveIngredient VARCHAR(100),
    TherapeuticClass VARCHAR(100),
    Manufacturer VARCHAR(100)
);

CREATE TABLE Stock (
    HID INT,
    MID INT,
    StockTimestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
    UnitPrice DECIMAL(10, 2) CHECK (UnitPrice >= 0),
    Qty INT DEFAULT 0 CHECK (Qty >= 0),
    ReorderLevel INT DEFAULT 10 CHECK (ReorderLevel >= 0),
    PRIMARY KEY (HID, MID, StockTimestamp),
    FOREIGN KEY (HID) REFERENCES Hospital(HID),
    FOREIGN KEY (MID) REFERENCES Medication(MID)
);

CREATE TABLE Prescription (
    PID INT PRIMARY KEY,
    CAID INT UNIQUE NOT NULL,
    DateIssued DATE NOT NULL,
    FOREIGN KEY (CAID) REFERENCES ClinicalActivity(CAID)
);

-- Many to many relationship between prescription and medication
   entities
CREATE TABLE Includes (
```

```
    PID INT,
    MID INT,
    Dosage VARCHAR(100),
    Duration VARCHAR(50),
    PRIMARY KEY (PID, MID),
    FOREIGN KEY (PID) REFERENCES Prescription(PID),
    FOREIGN KEY (MID) REFERENCES Medication(MID)
);

CREATE TABLE ContactLocation (
    CLID INT PRIMARY KEY,
    City VARCHAR(50),
    Province VARCHAR(50),
    Street VARCHAR(100),
    Number VARCHAR(10),
    PostalCode VARCHAR(10),
    PhoneLocation VARCHAR(15)
);

-- Many to many relationship between Patient and Contact Location
    entities
CREATE TABLE have (
    IID INT,
    CLID INT,
    PRIMARY KEY (IID, CLID),
    FOREIGN KEY (IID) REFERENCES Patient(IID),
    FOREIGN KEY (CLID) REFERENCES  ContactLocation(CLID)
);
```

4.2 -Altering tables

```
-- Altering patient table to add     email     attribute
ALTER TABLE Patient
ADD Email VARCHAR(100)
CHECK (Email LIKE '_%@_%._%');


-- Altering Staff table to add address column
ALTER TABLE Staff
ADD Address VARCHAR(50) ;
```

4.3 -Inserting data

```
-- Inserting sample rows
-- Patient
INSERT INTO Patient (IID, CIN, FullName, Birth, Sex, BloodGroup,
   Phone, Email)
VALUES
```

```
(1, 'CA123456', 'Youssef El Idrissi', '1985-04-12', 'M', 'A+', '0612345678', 'y.elidrissi@casablanca.ma'),
(2, 'RA789012', 'Fatima Zahra Bennani', '1990-09-25', 'F', 'O-', '0654321987', 'f.zahra@rabat.ma'),
(3, 'ME345678', 'Mohamed Amrani', '1978-01-30', 'M', 'B+', '0623456789', 'm.amrani@meknes.ma'),
(4, 'FE901234', 'Khadija El Fassi', '2000-06-15', 'F', 'AB+', '0678912345', 'k.elfassi@fes.ma'),
(5, 'MA567890', 'Rachid Toumi', '1995-03-10', 'M', 'O+', '0645789123', 'r.toumi@marrakech.ma');

-- Hospital
INSERT INTO Hospital (HID, Name, City, Region)
VALUES
(1, 'CHU Hassan II', 'Casablanca', 'Casablanca-Settat'),
(2, 'H pital Ibn Sina', 'Rabat', 'Rabat-Sal -K nitra'),
(3, 'H pital Mohammed V', 'Marrakech', 'Marrakesh-Safi'),
(4, 'H pital Regional Tanger', 'Tangier', 'Tanger-Tetouan-Al Hoceima'),
(5, 'Centre Hospitalier Universitaire F s', 'F s', 'F s-Mekn s');

-- Department
INSERT INTO Department (DEP_ID, HID, Name, Specialty) VALUES
(1, 1, 'Cardiology', 'Heart diseases'),
(2, 2, 'Neurology', 'Brain and nerves'),
(3, 3, 'Oncology', 'Cancer treatment'),
(4, 4, 'Pediatrics', 'Child healthcare'),
(5, 5, 'Orthopedics', 'Bone and muscles');


-- Staff
INSERT INTO Staff (STAFF_ID, FullName, Status, Address) VALUES
(1, 'Dr. Salma El Alami', 'Active', '12 Boulevard Mohammed V, Casablanca'),
(2, 'Dr. Yassine Najjar', 'Active', '45 Rue Al Qods, Rabat'),
(3, 'Dr. Amina Boulahlib', 'Active', '3 Jamaa el Fna Sq, Marrakech'),
(4, 'Dr. Rami Sedrati', 'Retired', '8 Medina St, Tangier'),
(5, 'Dr. Laila Chraibi', 'Active', '20 Andalus Ave, F s');

-- Workin
INSERT INTO Workin (STAFF_ID, DEP_ID)
VALUES
(1, 1),
(2, 2),
(3, 3),
(4, 4),
(5, 5);

-- ClinicalActivity
```

```sql
INSERT INTO ClinicalActivity (CAID, IID, STAFF_ID, DEP_ID, Date, Time)
VALUES
(1, 1, 1, 1, '2025-11-10', '09:30:00'),
(2, 2, 2, 2, '2025-11-11', '14:00:00'),
(3, 3, 3, 3, '2025-11-12', '10:15:00'),
(4, 4, 4, 4, '2025-11-13', '16:45:00'),
(5, 5, 5, 5, '2025-11-14', '08:00:00');

-- Appointment
INSERT INTO Appointment (CAID, Reason, Status)
VALUES
(1, 'Routine checkup', 'Completed'),
(2, 'Headache evaluation', 'Scheduled'),
(3, 'Cancer screening', 'Completed'),
(4, 'Vaccination', 'Cancelled'),
(5, 'Bone fracture follow-up', 'Completed');

-- Emergency
INSERT INTO Emergency (CAID, TriageLevel, Outcome)
VALUES
(1, 3, 'Discharged'),
(2, 2, 'Admitted'),
(3, 4, 'Transferred'),
(4, 1, 'Deceased'),
(5, 5, 'Discharged');

-- Insurance
INSERT INTO Insurance (INS_ID, Type)
VALUES
(1, 'CNOPS'),
(2, 'CNSS'),
(3, 'RAMED'),
(4, 'Private'),
(5, 'None');

-- Expense
INSERT INTO Expense (EXP_ID, INS_ID, CAID, Total)
VALUES
(1, 1, 1, 1200.50),
(2, 2, 2, 800.00),
(3, 3, 3, 1500.75),
(4, 4, 4, 500.25),
(5, 5, 5, 300.00);

-- Medication
INSERT INTO Medication (MID, Name, Form, Strength, ActiveIngredient, TherapeuticClass, Manufacturer)
VALUES
(1, 'Paracetamol', 'Tablet', '500mg', 'Paracetamol', 'Analgesic', 'Pharma Maroc'),
```

```
(2, 'Amoxicillin', 'Capsule', '250mg', 'Amoxicillin', 'Antibiotic', 'Sandoz Maroc'),
(3, 'Ibuprofen', 'Tablet', '400mg', 'Ibuprofen', 'Anti-inflammatory', 'Rameda'),
(4, 'Omeprazole', 'Capsule', '20mg', 'Omeprazole', 'Proton pump inhibitor', 'Medpharm Maroc'),
(5, 'Cetirizine', 'Tablet', '10mg', 'Cetirizine', 'Antihistamine', 'Pharma Plus');



-- Stock
INSERT INTO Stock (HID, MID, StockTimestamp, UnitPrice, Qty, ReorderLevel)
VALUES
(1, 1, '2025-11-01 08:00:00', 0.5, 100, 20),
(2, 2, '2025-11-02 09:00:00', 1.2, 50, 10),
(3, 3, '2025-11-03 10:00:00', 0.8, 75, 15),
(4, 4, '2025-11-04 11:00:00', 1.5, 30, 5),
(5, 5, '2025-11-05 12:00:00', 0.7, 90, 25);

-- Prescription
INSERT INTO Prescription (PID, CAID, DateIssued)
VALUES
(1, 1, '2025-11-10'),
(2, 2, '2025-11-11'),
(3, 3, '2025-11-12'),
(4, 4, '2025-11-13'),
(5, 5, '2025-11-14');

-- Includes
INSERT INTO Includes (PID, MID, Dosage, Duration)
VALUES
(1, 1, '500mg twice a day', '5 days'),
(2, 2, '250mg three times a day', '7 days'),
(3, 3, '400mg once a day', '3 days'),
(4, 4, '20mg once a day', '14 days'),
(5, 5, '10mg once a day', '10 days');

-- ContactLocation
INSERT INTO ContactLocation (CLID, City, Province, Street, Number, PostalCode, PhoneLocation) VALUES
(1, 'Casablanca', 'Casablanca-Settat', 'Mohammed V Blvd', '12', '20000', '0612345678'),
(2, 'Rabat', 'Rabat-Sal -K nitra', 'Al Qods St', '45', '10000', '0654321987'),
(3, 'Marrakech', 'Marrakesh-Safi', 'Jamaa el Fna Sq', '3', '40000', '0623456789'),
(4, 'Tangier', 'Tanger-Tetouan-Al Hoceima', 'Medina St', '8', '90000', '0678912345'),
```

```
(5, 'F s', 'F s-Mekn s', 'Andalus Ave', '20', '30000', '0645789123');

-- have
INSERT INTO have (IID, CLID)
VALUES
(1, 1),
(2, 2),
(3, 3),
(4, 4),
(5, 5);
```

4.4 -Updating data

```
-- Updating a patients phone number
UPDATE Patient
SET Phone = '0666114696'
WHERE IID = 5;
```

4.5 -Modifying data

```
-- Modifying a Hospitals region
UPDATE Hospital
SET City ='Settat',
    Region = 'Casablanca-Settat'
WHERE HID = 2;
```

4.6 -Deleting data

```
--Deleting a cancelled appointment
DELETE FROM Appointment
WHERE Status = 'Cancelled';
```

4.7 -Use of normalization

Normalization was applied to the MNHS database schema to ensure that it adheres to the Boyce-Codd Normal Form (BCNF). This process involved analyzing the functional dependencies within the database and decomposing relations that violated BCNF into smaller, well-structured relations. The goal was to eliminate redundancy, prevent update anomalies, and maintain data integrity. Each relation was carefully examined to ensure that every determinant is a candidate key, thereby achieving BCNF compliance.

4.8 -Queries script and execution

- QUERY 1:

```sql
SELECT FullName,
SUBSTRING_INDEX(FullName,'',-1) as last_name
FROM Patient
ORDER BY SUBSTRING_INDEX(FullNname,'',-1);
```

FullName	last_name
Fatima Zahra Bennani	Fatima Zahra Bennani
Khadija El Fassi	Khadija El Fassi
Mohamed Amrani	Mohamed Amrani
Rachid Toumi	Rachid Toumi
Youssef El Idrissi	Youssef El Idrissi

Figure 1: Query 1 Execution Result

- QUERY 2:

```sql
SELECT DISTINCT Type
FROM Insurance ;
```

Type
CNOPS
CNSS
RAMED
Private
None

Figure 2: Query 2 Execution Result

- QUERY 3:

```sql
SELECT S.Fullname
FROM Staff S, Hospital H , Departement D , Workin W
WHERE  S.STAFF_ID = W.STAFF_ID
AND W.DEP_ID = D.DEP_ID
AND D.HID = H.HID
AND H.City = 'Rabat';
```

FullName

Figure 3: Query 3 Execution Result

- QUERY 4:

```sql
SELECT DISTINCT A.CAID
FROM Appointment A
WHERE A.Status = 'Scheduled'
  AND EXISTS (
        SELECT Ca.CAID
        FROM ClinicalActivity Ca
        WHERE Ca.CAID = A.CAID
          AND Ca.Date <= CURDATE() + INTERVAL 7 DAY
  );
```

Figure 4: Query 4 Execution Result

- QUERY 5:

```sql
SELECT d.Name,d.DEP_ID
COUNT(a.CAID) AS nbr_apt
FROM Department d
JOIN ClinicalActivity ca ON ca.DEP_ID=d.DEP_ID
JOIN Appointment a ON a.CAID=ca.CAID
GROUP BY d.Name,d.DEP_ID;
```

Name	DEP_ID	nbr_apt
Cardiology	1	1
Neurology	2	1
Oncology	3	1
Orthopedics	5	1

Figure 5: Query 5 Execution Result

- QUERY 6:

```sql
SELECT HID,AVG(UnitPrice) as average_Unit_price
FROM Stock S
   NATURAL JOIN Hospital H
GROUP BY HID;
```

	HID	average_Unit_price
	1	0.500000
	2	1.200000
	3	0.800000
	4	1.500000
▶	5	0.700000

Figure 6: Query 6 Execution Result

- QUERY 7:

```sql
SELECT
    H.Name,
    COUNT(E.CAID) AS EmergencyCount
FROM
    Hospital H
JOIN
    Department D ON D.HID = H.HID
JOIN
    ClinicalActivity C ON C.DEP_ID = D.DEP_ID
JOIN
    Emergency E ON E.CAID = C.CAID
GROUP BY
    H.HID, H.Name
HAVING
    COUNT(E.CAID) > 20;
```

	Name	EmergencyCount

Figure 7: Query 7 Execution Result

- QUERY 8:

```sql
SELECT M.Name
FROM Medication M
WHERE M.TherapeuticClass = 'Antibiotic'
  AND M.Name IN (SELECT M2.Name
                 FROM Stock S , Medication M2
                 WHERE S.MID = M2.MID
                   AND S.UnitPrice <= 200);
```

	Name
▶	Amoxicillin

Figure 8: Query 8 Execution Result

- QUERY 9:

```sql
SELECT S.HID , S.MID
FROM Stock S
WHERE (
        SELECT COUNT(*)
        FROM Stock S2
        WHERE S2.HID = S.HID
          AND S2.UnitPrice > S.UnitPrice
      ) < 3;
```

	HID	MID
▶	1	1
	2	2
	3	3
	4	4
	5	5

Figure 9: Query 9 Execution Result

- QUERY 10:

```sql
SELECT d.Name,d.DEP_ID,
COUNT(CASE WHEN a.Status='Scheduled' THEN 1 END) AS
   nbr_scheduled,
COUNT(CASE WHEN a.Status='Completed' THEN 1 END) AS
   nbr_completed,
COUNT(CASE WHEN a.Status='Cancelled' THEN 1 END) AS
   nbr_cancelled
FROM Department d
JOIN ClinicalActivity ca ON ca.DEP_ID=d.DEP_ID
JOIN Appointment a ON a.CAID=ca.CAID
GROUP BY d.Name,d.DEP_ID;
```

Name	DEP_ID	nbr_scheduled	nbr_completed	nbr_cancelled
Cardiology	1	0	1	0
Neurology	2	1	0	0
Oncology	3	0	1	0
Orthopedics	5	0	1	0

Figure 10: Query 10 Execution Result

- QUERY 11:

```sql
SELECT *
FROM  Patient P
 NATURAL JOIN  Appointment A
 NATURAL JOIN ClinicalActivity CA
where CA.DATE >CURDATE() +INTERVAL 30 DAY;
```

IID	CAID	CIN	FullName	Birth	Sex	BloodGroup	Phone	Email	Reason	Status	STAFF_ID	DEP_ID	Date	Time

Figure 11: Query 11 Execution Result

- QUERY 12:

```sql
SELECT
    S.FullName,
    H.Name,
    COUNT(A.CAID) AS total_staff_appointement,
    (
        SELECT COUNT(A2.CAID)
        FROM Appointment A2
        JOIN ClinicalActivity CA2 ON A2.CAID = CA2.CAID
        JOIN Department D2 ON CA2.DEP_ID = D2.DEP_ID
        WHERE D2.HID = H.HID
    ) AS hospital_appointments_total,
    (
        COUNT(A.CAID) * 100.0 / (
            SELECT COUNT(A2.CAID)
            FROM Appointment A2
            JOIN ClinicalActivity CA2 ON A2.CAID = CA2.CAID
            JOIN Department D2 ON CA2.DEP_ID = D2.DEP_ID
            WHERE D2.HID = H.HID
        )
    ) AS percentage_share
FROM Staff S
JOIN ClinicalActivity CA ON S.STAFF_ID = CA.STAFF_ID
JOIN Appointment A ON CA.CAID = A.CAID
JOIN Department D ON CA.DEP_ID = D.DEP_ID
```

```
25 JOIN Hospital H ON D.HID = H.HID
26 GROUP BY S.STAFF_ID, S.FullName, H.HID, H.Name
27 ORDER BY H.Name, percentage_share DESC;
```

FullName	Name	total_staff_appointement	hospital_appointments_total	percentage_share
Dr. Laila Chraibi	Centre Hospitalier Universitaire Fès	1	1	100.00000
Dr. Salma El Alami	CHU Hassan II	1	1	100.00000
Dr. Yassine Najjar	Hôpital Ibn Sina	1	1	100.00000
Dr. Amina Boulahlib	Hôpital Mohammed V	1	1	100.00000

Figure 12: Query 12 Execution Result

- QUERY 13:

```
1 SELECT M.Name , H.Name
2 FROM Medication M , Hospital H
3 WHERE EXISTS (SELECT S.MID
4                FROM Stock S
5                WHERE S.mid = M.mid
6                  AND S.HID = H.HID
7                  AND S.Qty < S.ReorderLevel );
```

	Name	Name

Figure 13: Query 13 Execution Result

- QUERY 14:

```
1  SELECT S.HID
2  FROM Stock S
3  WHERE NOT EXISTS (
4      SELECT M.MID
5      FROM Medication M
6      WHERE M.TherapeuticClass = 'Antibiotic'
7      AND NOT EXISTS (
8          SELECT 1
9          FROM Stock S2
10         WHERE S2.HID = S.HID
11         AND S2.MID = M.MID
12     )
13 );
```

Figure 14: Query 14 Execution Result

- QUERY 15:

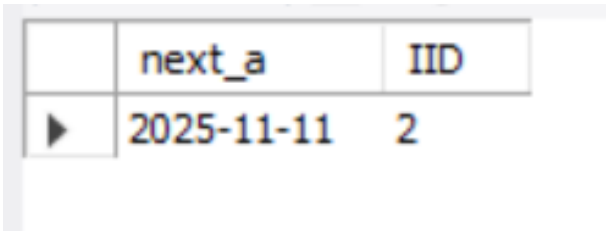
```sql
SELECT s.HID,m.TherapeuticClass,
 AVG(s.UnitPrice) AS avg_price,
 CASE
    WHEN AVG(s.UnitPrice)>c.avg_city THEN 'Above'
    WHEN AVG(s.UnitPrice)<c.avg_city THEN 'Below'
    WHEN AVG(s.UnitPrice)=c.avg_city THEN 'Equal'
 END AS avg_flag
FROM Medication m
JOIN Stock s ON m.MID=s.MID
JOIN (SELECT m.TherapeuticClass,
             AVG(UnitPrice) AS avg_city
             FROM Medication m
             JOIN Stock s ON m.MID=s.MID
             GROUP BY m.TherapeuticClass) c ON m.
                TherapeuticClass=c.TherapeuticClass
GROUP BY s.HID,m.TherapeuticClass,c.avg_city;
```

HID	TherapeuticClass	avg_price	avg_flag
1	Analgesic	0.500000	Equal
2	Antibiotic	1.200000	Equal
3	Anti-inflammatory	0.800000	Equal
4	Proton pump inhibitor	1.500000	Equal
5	Antihistamine	0.700000	Equal

Figure 15: Query 15 Execution Result

- QUERY 16:

```sql
SELECT MIN(Date) AS next_a, IID
FROM Patient
NATURAL JOIN Appointment
NATURAL JOIN ClinicalActivity
WHERE Status = 'Scheduled'
GROUP BY IID
ORDER BY next_a;
```



	next_a	IID
▶	2025-11-11	2

Figure 16: Query 16 Execution Result

- QUERY 17:

```sql
SELECT
    P.FullName,
    COUNT(E.CAID) AS total_visits,
    MAX(C.Date) AS recent_date
FROM
    Patient P
JOIN
    ClinicalActivity C ON P.IID = C.IID
JOIN
    Emergency E ON C.CAID = E.CAID
GROUP BY
    P.IID, P.FullName
HAVING
    COUNT(E.CAID) >= 2
    AND MAX(C.Date) >= '2025-11-01'
ORDER BY
    P.FullName;
```

	FullName	total_visits	recent_date

Figure 17: Query 17 Execution Result

- QUERY 18:

```sql
SELECT H.City ,
       H.Name ,
       COUNT(A.CAID) AS count,
       RANK() OVER (PARTITION BY H.City
       ORDER BY COUNT(A.CAID) DESC) AS rank
From Hospital H
INNER JOIN Department D ON H.HID = D.HID
INNER JOIN ClinicalActivity CA ON D.DEP_ID = CA.DEP_ID
INNER JOIN Appointment A ON A.CAID  = CA.CAID
WHERE A.Status = 'Completed'
  AND CA.DATE >= DATE_SUB(CURRENT_DATE,INTERVAL 90 DAY)
```

```
12 GROUP BY H.City , H.HID , H.Name
13 ORDER BY H.City , rank ;
```

City	Name	total_completed	rank_value
Casablanca	CHU Hassan II	1	1
Fès	Centre Hospitalier Universitaire Fès	1	1
Marrakech	Hôpital Mohammed V	1	1

Figure 18: Query 18 Execution Result

- QUERY 19:

```
1  SELECT S.MID, H.City
2  FROM Stock S
3  JOIN Hospital H ON S.HID = H.HID
4  WHERE S.MID IN (
5      SELECT S2.MID
6      FROM Stock S2
7      JOIN Hospital H2 ON S2.HID = H2.HID
8      GROUP BY S2.MID, H2.City
9      HAVING (MAX(S2.UnitPrice) - MIN(S2.UnitPrice)) / MIN(S2.
           UnitPrice) > 0.3
10 );
```

Result Grid

MID	City

Figure 19: Query 19 Execution Result

- QUERY 20:

```
1 SELECT *
2 FROM Stock s
3 WHERE Qty<0 OR UnitPrice<=0;
```

HID	MID	StockTimestamp	UnitPrice	Qty	ReorderLevel
NULL	NULL	NULL	NULL	NULL	NULL

Figure 20: Query 20 Execution Result

5 -Discussion

In this Lab , one of the main challenges we faced was ensuring that our database schema was properly normalized to eliminate redundancy and maintain data integrity. We had to carefully analyze the functional dependencies. Another challenge was writing complex SQL queries that involved multiple joins, subqueries, and aggregate functions to retrieve the required information accurately. We overcame these challenges by thoroughly reviewing database design principles and practicing SQL query writing and researching new functions and clauses that could be useful in the queries script. Additionally, we collaborated as a team to discuss and resolve any issues that arose during the implementation process.

6 -Conclusion

This lab provided us with experience in database design, normalization, and SQL query writing. We learned how to create a well structured database schema that adheres to normalization principles, ensuring data integrity and reducing redundancy. The process of writing and executing complex SQL queries helped us enhance our skills in data retrieval and manipulation. Overall, this lab has strengthened our understanding of database management systems and prepared us for more advanced projects.